Guided Lecture Notes

for

Proof, Set Theory, and Logic

MATH 330

Part II: Sets, Relations, and Functions

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These guided notes accompany the Fall 2026 revised course edition of *Elementary Set Theory: Proof, Set Theory, and Logic*. They are based on Richard P. Millspaugh's original textbook materials and lecture materials prepared and revised by Bryce A. Christopherson.

The blank spaces in the student copy are completed during class. Each blank occupies exactly the same space as the corresponding typed completion in the instructor copy, so the two packets have identical pagination. Definitions, theorem statements, and the surrounding explanations are provided so that class time can focus on examples, proof decisions, and the mathematical steps that connect them.

Contents of Part II

Chapter 3

Set Theory

3.1 What is a Set?

3.1.1 Naive Set Theory

Early work with sets often used *unrestricted comprehension*: any condition was assumed to determine a set of all objects satisfying that condition. Russell's paradox shows that this principle is inconsistent. Modern axiomatic set theories replace it with restricted principles, such as forming a subset of an already specified set. In this course we use informal set language, but every set-builder description should have a stated ambient set.

3.1.2 Russell's Paradox

Example 3.1.

The Barber's Paradox. A certain small town has one barber, who is a man and shaves exactly those men in town who do not shave themselves. Does the barber shave himself?

Russell considered this sort of thing more formally by constructing the (purported) set

$$R = \{x \mid x \notin x\}.$$

If $R \in R$, then its defining condition gives $R \notin R$; if $R \notin R$, then its defining condition gives $R \in R$. The lesson is not merely that self-membership should be forbidden. Rather, the collection of all objects satisfying an arbitrary condition need not be a set. Standard axiomatic systems avoid the contradiction by restricting comprehension, as described below.

3.1.3 A Glimpse of ZFC

The most commonly used formal foundation for ordinary mathematics is *Zermelo–Fraenkel set theory with the Axiom of Choice*, usually abbreviated *ZFC*. It is not the only possible foundation, but it is the standard one. ZFC uses first-order logic with equality and one basic relation, membership $\in$. Every variable in its formal language ranges over sets. Numbers, ordered pairs, functions, and other mathematical objects can all be represented using sets, although we will continue to treat them as the familiar objects that they are. For example, one common set-theoretic representation of an ordered pair is

$$(a, b) = \big\{\{a\}, \{a, b\}\big\}.$$

The *Axiom of Extensionality* formalizes our criterion for equality of sets:

$$\forall A\, \forall B \left[\big(\forall x\, (x \in A \Leftrightarrow x \in B)\big) \Rightarrow A = B\right].$$

In words, if two sets have exactly the same elements, then they are equal. Consequently, a set is completely determined by its elements.

The principle that most directly addresses Russell's paradox is the *Separation schema*. For every property $\varphi(x)$ that can be expressed in the language of set theory, Separation supplies an axiom of the form

$$\forall A\, \exists B\, \forall x \left[x \in B \Leftrightarrow \big(x \in A \wedge \varphi(x)\big)\right].$$

Thus, from an already existing set A, we may form the subset

$$B = \{x \in A \mid \varphi(x)\}.$$

The word *schema* means that this is actually a pattern giving one axiom for each allowable formula φ. Separation does not say that $\{x \mid \varphi(x)\}$ exists without an ambient set. In particular, it does not provide Russell's purported set $\{x \mid x \notin x\}$.

The other axioms guarantee the existence of the sets needed for familiar constructions. Pairing gives sets such as $\{a, b\}$; Union and Power Set justify the formation of unions and power sets; and Infinity guarantees the existence of an infinite set. The Replacement schema allows the outputs of a definable functional rule, when applied to the elements of a set, to be collected into another set. Foundation rules out membership loops such as $x \in x$. Finally, the Axiom of Choice says that, given a set of nonempty sets, there is a function choosing one element from each of them. ZF consists of the preceding set-theoretic principles without Choice; ZFC means ZF together with Choice. Equivalent presentations sometimes package a few of these principles differently.

These axioms are what justify the constructions that we use informally. For example, Pairing guarantees that $\{A, B\}$ exists, and the Union axiom then guarantees the existence of $A \cup B$. The Power Set axiom guarantees the existence of $\mathcal{P}(A)$. A definition describes the intended set, while the axioms ensure that such a set actually exists.

We will not derive our results directly from the axioms of ZFC. The purpose of this discussion is only to show the formal framework underneath our informal work. In particular, ZFC has no set containing every set. When we later use a universal set U, it will mean a fixed ambient set for the problem under consideration, not a set containing all mathematical objects.

3.2 Elements and Subsets

3.2.1 Elements

We will think of a set as a collection of objects, called its *elements*. These objects can be any objects we are interested in discussing. Any roster or description must determine membership unambiguously. In particular, a set-builder description should specify an ambient set and a condition that determines membership. We consider two sets equal if they contain exactly the same elements. An object is either an element of a set or it is not; elements have no order and repetition does not create a new element.

One way to indicate a set is to simply list all of the elements between set braces; i.e., between the symbols $\{$ and $\}$. The set of positive integers less than 3 is $\{1, 2\}$. We will find it useful to have a special notation for the set which has no elements. To this end, we let the symbol $\varnothing$ denote the set $\{\}$ (i.e. $\varnothing = \{\}$).

Example 3.2.

Example 3.3.

We indicate that the object x is an element of the set A by writing $x \in A$. We write $x \notin A$ if x is not an element of A.

Listing all elements works well for a small finite set. For larger sets we use *set-builder notation*

$$\{x \in U \mid P(x)\},$$

read "the set of all x in U such that $P(x)$." The ambient set U is part of the definition. For example, $\{x \in \mathbb{R} \mid x^2 - 2 = 0\}$ is a two-element set, whereas $\{x \in \mathbb{Q} \mid x^2 - 2 = 0\} = \varnothing$.

There are also a few sets that are useful enough to deserve their own special notation. In particular, $\mathbb{N}$ denotes the set of natural numbers, $\mathbb{Z}$ the set of integers, $\mathbb{Q}$ the set of rational numbers, $\mathbb{R}$ the set of real numbers, and $\mathbb{C}$ the set of complex numbers.

3.2.2 Subsets

Definition 3.4.
Given two sets A and B, we say that A is a *subset* of B, or that A is contained in B, if every element of A is also an element of B. We write $A \subseteq B$.

Since every natural number is also an integer and every integer is a real number, $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{R}$.

Example 3.5.

Given any set A, it is certainly true that every element of A is an element of A. In other words, every set is a subset of itself. Note also that the empty set is a subset of every set A since there are no elements of the empty set which are not in A.

Definition 3.6.
A subset B of A is a *proper subset* of A if $B \neq A$. We write $B \subsetneq A$. In particular, $\emptyset \subsetneq A$ whenever A is nonempty.

Theorem 3.7.
If $A \subseteq B$ *and* $B \subseteq C$, *then* $A \subseteq C$.

Proof.

3.2.3 Universal Sets

In many instances, all of the sets we may be interested in are subsets of some particular set U. In this case we say that U is a *universal set*, or that U is the *universe*. For example, all of the functions we study in single variable calculus have domains and ranges that are subsets of the universal set $\mathbb{R}$.

3.2.4 Equality of Sets

We say that two sets are equal if they contain exactly the same elements. In other words, $A = B$ means that every element of A is an element of B and every element of B is an element of A. A good test for this is the following one:

Theorem 3.8.
Given any two sets A *and* B, $A = B$ *if and only if* $A \subseteq B$ *and* $B \subseteq A$.

This theorem will become one of our most valuable tools for showing that two sets are equal.

3.3 Operations on Sets

3.3.1 Union and Intersection

Definition 3.9.
The *union* of two sets A and B is the set:

$$A \cup B = \{x \mid (x \in A) \vee (x \in B)\}$$

Definition 3.10.
The *intersection* of A and B is the set:

$$A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}$$

Example 3.11.

Theorem 3.12.
Let A, B, *and* C *be any sets in the universal set* U. *Then:*

1. *(Commutative Laws)*
   (a) $A \cup B = B \cup A$
   (b) $A \cap B = B \cap A$
2. *(Associative Laws)*
   (a) $A \cup (B \cup C) = (A \cup B) \cup C$
   (b) $A \cap (B \cap C) = (A \cap B) \cap C$
3. *(Distributive Laws)*
   (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. *(Idempotence)*
   (a) $A \cup A = A$
   (b) $A \cap A = A$
5. *(Identities)*
   (a) $A \cup \varnothing = A$
   (b) $A \cap U = A$

Proof.

3.3.2 Complements

Definition 3.13.
The *relative complement* of B in A (also called the set difference) is the set:

$$A \setminus B = \{a \mid a \in A \text{ and } a \notin B\}$$

In a given universe U, we may also define the *complement* of a set A to be the set:

$$A' = U \setminus A$$

Theorem 3.14 (De Morgan's Laws).
Let A, B, *and* C *be sets. Then:*

(i) $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$, *and*

(ii) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

Proofs.

3.3.3 Cartesian Products

Definition 3.15.
The *(Cartesian) product* of two sets A and B is the set:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\},$$

where (a, b) denotes an ordered pair, not an interval.

Ordered pairs satisfy $(a, b) = (a', b')$ if and only if $a = a'$ and $b = b'$.

Note that $A \times B$ is simply the set of all ordered pairs with first coordinates in A and second coordinates in B. For example, the Cartesian plane used in Calculus is the set $\mathbb{R} \times \mathbb{R}$.

Theorem 3.16.
For any sets A, B, *and* C:

(i) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

(ii) $(A \cap B) \times C = (A \times C) \cap (B \times C)$

(iii) $(A \setminus B) \times C = (A \times C) \setminus (B \times C)$

Proof.

The products $A \times B$ and $B \times A$ need not be equal because their coordinates play different roles. They are equal in special cases, for example when $A = B$ or when at least one factor is empty. The proof of the following theorem involves making obvious changes to the previous proof and checking that all of the details still work.

Theorem 3.17.
For any sets A, B, *and* C:

(i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

(iii) $A \times (B \smallsetminus C) = (A \times B) \smallsetminus (A \times C)$

It may be tempting to assume that we could combine these two theorems somehow and obtain statements like $(A \times B) \smallsetminus (C \times D) = (A \smallsetminus C) \times (B \smallsetminus D)$. This statement is generally false, however.

Example 3.18.

3.4 Collections of Sets

Some of the structures used in pure mathematics require the use of sets whose elements are other sets. It is customary, though not necessary, to refer to these kinds of sets as collections of sets.

3.4.1 The Power Set of a Set

Definition 3.19.
Let A be any set. The *power set* of A is $\mathcal{P}(A) = \{B \mid B \subseteq A\}$.

In words, the power set of A is the set whose elements are the subsets of A. Note that for any set A we have $\varnothing \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$, so $\mathcal{P}(A)$ is nonempty for every set A. The power set of A is frequently denoted 2^A.

Theorem 3.20.
For any sets A *and* B, $A \subseteq B$ *if and only if* $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Proof.

Theorem 3.21.
For any sets A *and* B, $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Proof.

Chapter 4

Relations

Introduction

The first mathematical relation most students become aware of is the standard order on the natural numbers, which is later extended to the integers and finally the reals. A relation is a much more general concept, and by now you have probably worked with several others, perhaps without thinking about them as relations. In this chapter we define what we mean by a relation between sets, then define an equivalence relation on a set and use that idea to develop the set of rational numbers.

4.1 Relations

Definition 4.1.
Given two sets A and B, a *relation* from A to B is a subset of $A \times B$. The relations we will be most interested in are usually from a set A to itself, in which case we say that the relation is a relation on A. If S is a relation, we frequently use the notation xSy to indicate that the ordered pair (x, y) is in S.

Let's look at some examples of relations.

Example 4.2.

Example 4.3.

Example 4.4.

Definition 4.5.
Let A be a set and let $\sim$ be a relation on A. We say that $\sim$ is:

1. *reflexive* if $a \sim a$ for every $a \in A$;
2. *symmetric* if $a \sim b$ implies $b \sim a$ for every $a, b \in A$;
3. *antisymmetric* if $a \sim b$ and $b \sim a$ imply $a = b$ for every $a, b \in A$.
4. *transitive* if $a \sim b$ and $b \sim c$ imply $a \sim c$ for every $a, b, c \in A$.

Note that symmetry and antisymmetry are not logical opposites, though the names may lead you to believe otherwise. It is possible for a relation to satisfy both of these properties, or to satisfy neither of them.

Example 4.6.

4.2 Partial Orders

Definition 4.7. A relation $\preceq$ on a set A is a *partial order* if it is reflexive, antisymmetric, and transitive. The pair $(A, \preceq)$ is a *partially ordered set*, or *poset*.

Two elements $a, b \in A$ are *comparable* if $a \preceq b$ or $b \preceq a$. A partial order in which every pair of elements is comparable is called a *total order*.

Example 4.8.

Example 4.9.

4.3 Equivalence Relations

Consider the following question: Are $\pi/4$ and $25\pi/4$ measurements of the same angle? It's certainly true that when we place these angles in standard position they have the same terminal side, which means that we can treat them as the same much of the time. On the other hand, if we think of the angle as a rotation, these two angles are certainly different. If you're playing pin the tail on the donkey, being spun through an angle of $25\pi/4$ is going to make you dizzier than being spun through an angle of $\pi/4$. Intuitively, the terminal side of the angle in this setting tells us what direction we end up pointing in. The angle tells us how far we rotated to end up pointing in that direction. When we are only concerned with the terminal side of an angle, how do we tell when two numbers are measures for *equivalent* angles? You probably learned in a trigonometry or precalculus class that two measurements represent equivalent angles when they differ by an integer multiple of 2π. In other words, two angles α and β are said to be *equivalent* when there is an integer n such that $\alpha - \beta = 2\pi n$. In the following example we consider this relation between real numbers.

Example 4.10.

Definition 4.11.
A relation on a set A that is reflexive, symmetric, and transitive is said to be an *equivalence relation* on A.

4.3.1 Equivalence Classes

Definition 4.12.
Let A be a set and let $\sim$ be an equivalence relation on A. For any $a \in A$ the set $[a] = \{b \in A \mid b \sim a\}$ is called the *equivalence class* of a.

Given a set A and an equivalence relation on A, the set of equivalence classes forms a *partition* of A. In other words, the equivalence classes are nonempty subsets of A, every element of A belongs to an equivalence class, and no two distinct equivalence classes intersect. We formalize this in the following theorem.

Theorem 4.13.
Let A *be a nonempty set and* $\sim$ *an equivalence relation on* A*. For each* $a \in A$*, let*

$$[a] = \{b \in A \mid b \sim a\}.$$

(i) For each $a \in A$, $a \in [a]$.

(ii) If $[a] \neq [b]$*, then* $[a] \cap [b] = \varnothing$.

Proof.

Example 4.14.

In the previous example, note that we had several ways to refer to each equivalence class. For example $[1] = [16]$, so we may as well have used $[16]$ as a name for this equivalence class. In this context the numbers 1 and 16 (or any other member of the class) are called *representatives* of this equivalence class, and any representative can be used to name the equivalence class.

4.4 The Rational Numbers

In this section we use equivalence classes to construct the rational numbers. Assume that $\mathbb{N}$ and $\mathbb{Z}$ have their usual properties and recall that $0 \notin \mathbb{N}$ in this text. Begin with

$$\mathbb{F} = \mathbb{Z} \times \mathbb{N}.$$

Its elements are ordered pairs (a, b), not yet numerical quotients. We use a/b as a convenient *formal symbol* for the pair (a, b); the condition $b \in \mathbb{N}$ guarantees that its second coordinate is nonzero.

Note that a fraction is not quite the same thing as a rational number, for example $1/2$ and $3/6$ are different fractions that represent the same rational number. We define an equivalence relation $\cong$ on $\mathbb{F}$ that clarifies when two fractions represent the same rational as follows:

Definition 4.15.
Given two fractions a/b and c/d in $\mathbb{F}$, we say that $a/b \cong c/d$ if $ad = bc$.

Theorem 4.16.
The relation $\cong$ is an equivalence relation on $\mathbb{F}$.

Proof.

We now define the set of rationals to be the set of equivalence classes of fractions. In this construction the two fractions 1/2 and 3/6 do indeed represent the same rational number since they are representatives of the same equivalence class. This construction might be a bit unsatisfying in the sense that it tells us nothing about how we might add or multiply rational numbers. Even if we know how to add fractions, how would we add two equivalence classes of fractions? One possibility is that we might add two equivalence classes by adding their representatives, so for example we might try:

$$\left[\frac{1}{2}\right] \oplus \left[\frac{2}{3}\right] = \left[\frac{1(3)+2(2)}{2(3)}\right] = \left[\frac{5}{6}\right]$$

There is a potential problem with this, though. There are infinitely many choices of fractions representing each rational number. Are we sure that we arrive at the same result for all of those possible choices? The next theorem says that we do.

Theorem 4.17.
Suppose that $a/b \cong x/y$ and $c/d \cong w/z$, where a/b, c/d, x/y, and w/z are all in $\mathbb{F}$, then:

$$\frac{ad+bc}{bd} \cong \frac{xz+yw}{yz}.$$

Proof.

The proof of the following theorem is left as an exercise.

Theorem 4.18.
Suppose that $a/b \cong x/y$ and $c/d \cong w/z$, where a/b, c/d, x/y, and w/z are all in $\mathbb{F}$, then:

$$\frac{ac}{bd} \cong \frac{xw}{yz}.$$

Theorem 4.17 and Theorem 4.18 allow us to make the following definitions, where we may choose any representative from each equivalence class.

Definition 4.19.
Let $[a/b]$ and $[c/d]$ be rational numbers, then we define addition and multiplication by:

$$\left[\frac{a}{b}\right] \oplus \left[\frac{c}{d}\right] = \left[\frac{ad+bc}{bd}\right] \quad \text{and} \quad \left[\frac{a}{b}\right] \otimes \left[\frac{c}{d}\right] = \left[\frac{ac}{bd}\right].$$

Chapter 5

Functions

Introduction

As anybody who has taken a calculus class knows, functions are important in mathematics. In this chapter we will define a function between two sets as a special type of relation between those sets. This set-theoretic definition emerged from nineteenth- and early twentieth-century efforts to make the notions of function, relation, and set mathematically precise. These efforts became especially important once paradoxes exposed limitations in informal set reasoning.

5.1 Definition

You probably think of functions in several ways based on what you have seen in previous courses. Functions can be rules for computing things, some people think of them as machines (plug in one number and get out another), and of course we all know that a graph is only the graph of a function if it passes the vertical line test. Most of these ideas match the way that mathematicians thought of functions (and the way they worked with them) at some time or another. It wasn't until the late 19th century that the concept of a function was defined in terms of sets. We present such a definition here:

Definition 5.1.
Let A and B be sets. A *function* f from A to B is a relation from A to B with the additional property that for each $a \in A$ there is exactly one ordered pair (a, b) in f having a as a first coordinate. In this case, the set A is called the *domain* of f and the set B is called the *codomain* of f. If f is a function from A to B, we denote this by writing $f : A \to B$.

Do not confuse the codomain of a function with its range. The codomain of a function is the set that second coordinates must come from. The range is the subset of the codomain containing all of those second coordinates. Sometimes those sets are the same, but sometimes they are not.

Note that our definition requires that every element of the domain be the first coordinate of one, and only one, ordered pair in a function. It does not, however, require that each element of the codomain be a second coordinate, or that it be the second coordinate of only one ordered pair. Satisfying these requirements would make our function surjective or injective, respectively. We discuss these kinds of functions in Section 5.2.

Before going any further, let's consider a couple of examples.

Example 5.2.

The functions in the previous example are obviously constructed to fit the definition, but may not strike you as the kinds of things you think of as functions. What about the kinds of functions we are used to?

Example 5.3.

It probably seems unnatural to you to write the squaring function $f : \mathbb{R} \to \mathbb{R}$ as we did in Example 5.3. Wouldn't it be easier to write this function as $f(x) = x^2$, the same way we did in algebra or calculus classes? Once we know that f is a function, we can use this notation.

Suppose that $f : A \to B$ is a function. If $a \in A$ and $(a, b) \in f$, we say that b is the *value* of the function f at a and denote this by writing $b = f(a)$.

Now the notation $f(x) = x^2$ indicates that for each $x \in \mathbb{R}$, x^2 is the value of the function at x. Please be careful to distinguish between the name of the function f and an arbitrary value of the function $f(x)$.

Example 5.4.

5.1.1 Images and Preimages

Definition 5.5. Let $f : A \to B$ be a function.

(i) If $S \subseteq A$, the *image* of S under f is

$$f[S] = \{f(s) \mid s \in S\} \subseteq B.$$

In particular, the range of f is $f[A]$.

(ii) If $T \subseteq B$, the *preimage* of T under f is

$$f^{-1}[T] = \{a \in A \mid f(a) \in T\}.$$

The notation $f^{-1}[T]$ denotes a set and is meaningful for *every* function f, whether or not f has an inverse function. It should not be confused with the value $f^{-1}(y)$ of an inverse function. For example, if $f : \mathbb{R} \to \mathbb{R}$ is $f(x) = x^2$, then

$$f^{-1}[\{4\}] = \{-2, 2\},$$

even though f has no inverse function from $\mathbb{R}$ to $\mathbb{R}$.

Images and preimages interact differently with set operations. For $S_1, S_2 \subseteq A$ and $T_1, T_2 \subseteq B$,

$$\begin{aligned} f[S_1 \cup S_2] &= f[S_1] \cup f[S_2], \\ f^{-1}[T_1 \cup T_2] &= f^{-1}[T_1] \cup f^{-1}[T_2], \\ f^{-1}[T_1 \cap T_2] &= f^{-1}[T_1] \cap f^{-1}[T_2], \\ f^{-1}[B \setminus T_1] &= A \setminus f^{-1}[T_1]. \end{aligned}$$

In general only $f[S_1 \cap S_2] \subseteq f[S_1] \cap f[S_2]$; equality holds whenever f is injective.

5.1.2 Binary Operations

Standard addition and multiplication on $\mathbb{R}$ are both examples of functions used to compute a single real number from two given real numbers. This kind of function is important enough to deserve its own name.

Definition 5.6.
A *binary operation* on a set X is a function $* : X \times X \to X$. If $*$ is a binary operation on X, we usually use the notation $x * y$ to denote the value $*(x, y)$.

Example 5.7.

Example 5.8.

Definition 5.9. We say that a binary operation $*$ on a set X is:

- *commutative* if $x * y = y * x$ for all $x, y \in X$.
- *associative* if $(x * y) * z = x * (y * z)$ for all $x, y, z \in X$.

Addition and multiplication on $\mathbb{Z}$ (or on $\mathbb{N}$ or $\mathbb{R}$ for that matter) are commutative and associative, which you have probably known since your first algebra class. Subtraction on $\mathbb{Z}$ is not commutative since, for example, $4 - 2 \neq 2 - 4$. Subtraction on $\mathbb{Z}$ also fails to be associative since, for example, $(5 - 1) - 3 = 1 \neq 7 = 5 - (1 - 3)$.

Example 5.10.

5.2 Injective, Surjective, and Bijective Functions

Definition 5.11.
A function $f : X \to Y$ is said to be:

- *injective* or *one-to-one* if for all $x, y \in X$, if $f(x) = f(y)$, then $x = y$.
- *surjective* or *onto* if for each $y \in Y$, there is an $x \in X$ such that $f(x) = y$.
- *bijective* or *a one-to-one correspondence* if f is both injective and surjective.

Example 5.12. Classify the functions from Example 5.4 as injective, surjective, both, or neither, under the assumptions stated below.

(i) Let A be nonempty and consider the identity function $i : A \to A$.

(ii) Assume that A and B each have more than one element and let $k : A \to B$ be the constant function $k(a) = b_0$.

(iii) Assume that A and B each have more than one element and consider the coordinate projections $\pi_A : A \times B \to A$ and $\pi_B : A \times B \to B$.

(iv) Consider the characteristic function $\chi_A : \mathbb{R} \to \{0, 1\}$.

5.3 Compositions of Functions

Definition 5.13.
Let $f : X \to Y$ and $g : Y \to Z$ be functions. The *composition* is the function from X to Z defined by $(g \circ f)(x) = g(f(x))$. In other words, to find $(g \circ f)(x)$, first find $f(x)$, then plug the result into the function g. In terms of ordered pairs, the composition is

$$g \circ f = \{(x, z) \mid (x, y) \in f \text{ and } (y, z) \in g \text{ for some } y \in Y\}.$$

Theorem 5.14. *Let $f : X \to Y$ and $g : Y \to Z$ be functions.*

(i) If f and g are injective, then $g \circ f : X \to Z$ is injective.

(ii) If f and g are surjective, then $g \circ f : X \to Z$ is surjective.

Proof.

Combining the two parts of Theorem 5.14, we have the following:

Corollary 5.15.
If $f : X \to Y$ and $g : Y \to Z$ are bijective functions, then $g \circ f : X \to Z$ is bijective.

It is natural to ask whether or not the converses of the statements in Theorem 5.14 are true. We consider the converse of Theorem 5.14 *(i)* in the following example and theorem.

Example 5.16.

This example shows that it is possible for $g \circ f$ to be injective when g is not. The next result says that if $g \circ f$ is injective, then it does follow that f is injective.

Theorem 5.17.
Let $f : X \to Y$ and $g : Y \to Z$ be functions such that $g \circ f : X \to Z$ is injective. Then $f : X \to Y$ must be injective.

Proof.

We leave the proof of the corresponding result for surjective functions as an exercise.

Theorem 5.18.
Let $f : X \to Y$ and $g : Y \to Z$ be functions such that $g \circ f : X \to Z$ is surjective. Then $g : Y \to Z$ must be surjective.

5.3.1 Inverse Relations and Inverse Functions

Definition 5.19.
If $f : X \to Y$ is a function, the *inverse relation* of f is the relation g from Y to X given by

$$g = \{(y, x) \mid (x, y) \in f\}.$$

Every function has an inverse relation. The inverse relation is not generally a function, as the following example shows. It is also different from the preimage operation $T \mapsto f^{-1}[T]$ introduced in Section 5.1.1.

Example 5.20.

Definition 5.21.
If $f : X \to Y$ is a function and its inverse relation is also a function, we say that f is *invertible* and use $f^{-1} : Y \to X$ to denote the resulting *inverse function*.

Question 5.22.
When is a function $f : X \to Y$ invertible?

A real-valued function that passes the "horizontal line test" has an inverse function defined on its range. For a function $f : \mathbb{R} \to \mathbb{R}$ to have an inverse function whose domain is all of $\mathbb{R}$, it must also be surjective. One way to understand the horizontal line test is that the graph of an inverse relation is the reflection of the original graph through the line $y = x$. Since we want this reflection to pass the vertical line test, no horizontal line should meet the original graph more than once. This is the injectivity part of invertibility.

Thus the horizontal-line test captures injectivity, but invertibility also depends on the stated codomain. We begin by proving that every invertible function is injective.

Theorem 5.23.
If $f : X \to Y$ is an invertible function, then f is injective.

Proof.

Unfortunately, this is not a complete answer to our question because the converse of Theorem 5.23 is not true. We have found a condition that all invertible functions must satisfy, but not all functions that satisfy this condition are invertible. The following example illustrates the difficulty.

Example 5.24.

For a function $f : X \to Y$ to be invertible, every element of the codomain Y must be a first coordinate of some ordered pair in the inverse relation. That in turn means that every element of the codomain must be the second coordinate of some ordered pair in f. This is exactly what it means to say that f is surjective. This leads us to believe the following:

Theorem 5.25.
If $f : X \to Y$ is invertible, then f is surjective.

Proof.

We are now ready to give a complete answer to our question in the form of the following theorem.

Theorem 5.26.
A function $f : X \to Y$ is invertible if and only if it is bijective.

Proof.